

VIVEKANANDA INSTITUTE OF PROFESSIONAL STUDIES - TECHNICAL CAMPUS
Grade A++ Accredited Institution by NAAC
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An ISO 9001:2015 Certified

SCHOOL OF ENGINEERING & TECHNOLOGY

B.Tech Programme: Artificial Intelligence and Machine Learning – (A) (7th Semester)

Course Title: Advances in Machine Learning Lab

Course Code: AIML411P

Submitted By:
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Enrolment No: 01117711623

VISION OF INSTITUTE

To be an educational institute that empowers the field of engineering to build a sustainable future by providing quality education with innovative practices that supports people, planet and profit.

MISSION OF INSTITUTE

To groom the future engineers by providing value-based education and awakening students' curiosity, nurturing creativity and building capabilities to enable them to make significant contributions to the world.

S. No.	Experiment	Date	Marks (Total Ten) which includes (Laboratory Assessment, Class participation and Viva)	Remarks	Updated Marks	Faculty Signature
1.	Implement a deep neural network from scratch using TensorFlow or PyTorch, gaining hands-on experience in building complex neural architectures.					
2.	Utilize pre-trained models and perform transfer learning to solve real-world problems efficiently.					
3.	Implement a GAN to generate synthetic data and explore its applications in image generation and data augmentation.					
4.	Apply NLP techniques to process and analyze textual data, including sentiment analysis and named entity recognition.					
5.	Build RL agents and train them using OpenAI Gym or Stable Baselines to solve challenging tasks.					
6.	Understand the interpretability of ML models by using LIME or SHAP to explain model predictions.					

7.	Use AutoML and Hyperparameter tuning tools to automate the model selection and optimization process.					
8.	Analyze time series data, perform forecasting, and evaluate model performance using Prophet or stats-models.					
9.	Compress and quantize large ML models to make them suitable for deployment on resource constrained devices.					
10.	Explore federated learning concepts and implement distributed ML models using TensorFlow Federated.					
11.	Generate adversarial attacks on ML models and implement defense mechanisms to enhance model robustness.					
12.	Utilize Ray Tune to perform hyperparameter search and optimize ML models efficiently.					